



Aror University of Art, Architecture, Design & Heritage Sukkur

BS Artificial Intelligence (Section-D)

Class: BS AI

Discrete Structures

Term: Spring 2026

Due Date: 12-02-2026

Assignment # 01

Max Marks: 5+5

Instructor: Sarfaraz Ahmed Baloch

Date: 04-02-2026

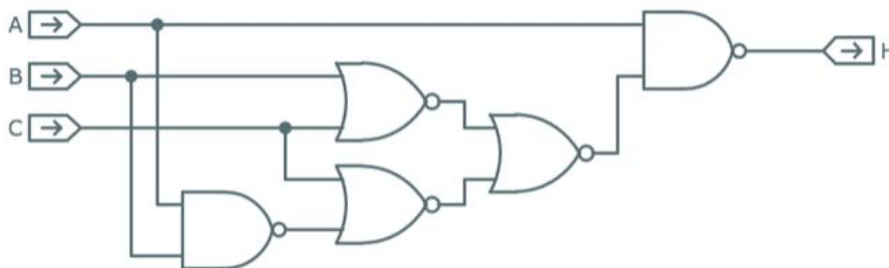
Group A

Q#01 Explain in detail why Discrete Mathematics is necessary to learn for the students of AI?

Q#02

- (i) What is Propositional Logic and why it is necessary for AI students to understand Propositional Logic?
- (ii) Give two examples related to the application of AI for each of the following propositions: Conjunction, Disjunctions (Inclusive or and Exclusive or), Implication and Bi-implication.

Q#03 Consider the Boolean function $H(A, B, C)$ shown in diagram.



- (i) Give output expression of every gate on the diagram.
- (ii) Write the truth table of the above diagram.

Q#04

- (i) Explain through examples why conditional statement and its converse are not equivalent.
- (ii) Explain through examples why conditional statement and its inverse are not equivalent.

- (iii) Explain through examples how conditional statement and its contrapositive are equivalent.

Q#05 Let p , q , and r be the following propositions:

p : You get an A on the final exam; q : You do every exercise in this book; r : You get an A in this class.

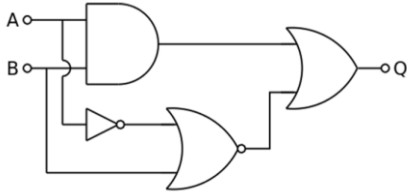
Write these propositions using p , q , and r and logical connectives.

- a) You get an A in this class, but you do not do every exercise in this book.
- b) You get an A on the final, you do every exercise in this book, and you get an A in this class.
- c) To get an A in this class, it is necessary for you to get an A on the final.
- d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.
- e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.
- f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

Group B

Q#01

- (i) Write down the Boolean expression for each output of every gate in the following logic circuit.



- (ii) Write the truth table of the above logic circuit.

Q#02 Construct a truth table for each of these compound propositions.

- a) $p \rightarrow (\neg q \vee r)$
- b) $\neg p \rightarrow (q \rightarrow r)$
- c) $(p \rightarrow q) \vee (\neg p \rightarrow r)$
- d) $(p \rightarrow q) \wedge (\neg p \rightarrow r)$
- e) $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$
- f) $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$

Q#03 What is the negation of each of these propositions?

- a) Mei has an MP3 player.
- b) There is no pollution in New Jersey.
- c) $2 + 1 = 3$.
- d) The summer in Maine is hot and sunny.
- e) Jennifer and Teja are friends.
- f) There are 13 items in a baker's dozen.
- g) Abby sent more than 100 text messages every day.
- h) 121 is a perfect square.
- i) Steve has more than 100 GB free disk space on his laptop.
- j) Zach blocks e-mails and texts from Jennifer.
- k) $7 \cdot 11 \cdot 13 = 999$.

l) Diane rode her bicycle 100 miles on Sunday.

Q#04 The truth value of the negation of a proposition in fuzzy logic is 1 minus the truth value of the proposition. What are the truth values of the statements “Fred is not happy” and “John is not happy?”

Q#05 Show that each of these conditional statements is a tautology by using truth tables.

a) $(p \wedge q) \rightarrow p$

b) $\neg p \rightarrow (p \rightarrow q)$

c) $\neg (p \rightarrow q) \rightarrow p$

d) $[\neg p \wedge (p \vee q)] \rightarrow q$

e) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

Q#01 Let p and q be the propositions “The election is decided” and “The votes have been counted,” respectively. Express each of these compound propositions as an English sentence.

- a) $\neg p$
- b) $p \vee q$
- c) $\neg p \wedge q$
- d) $q \rightarrow p$
- e) $\neg q \rightarrow \neg p$
- f) $\neg p \rightarrow \neg q$
- g) $p \leftrightarrow q$
- h) $\neg q \vee (\neg p \wedge q)$

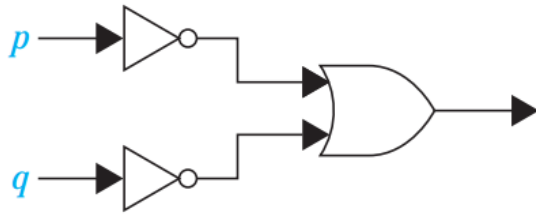
Q#02 Construct a truth table for each of these compound propositions.

- a) $(p \vee q) \rightarrow (p \oplus q)$
- b) $(p \oplus q) \rightarrow (p \wedge q)$
- c) $(p \vee q) \oplus (p \wedge q)$
- d) $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$
- e) $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$
- f) $(p \oplus q) \rightarrow (p \oplus \neg q)$

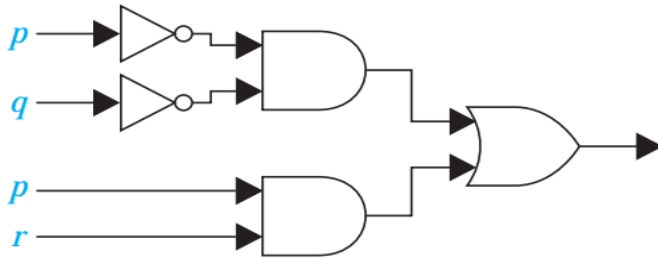
Q#03 The truth value of the conjunction of two propositions in fuzzy logic is the minimum of the truth values of the two propositions. What are the truth values of the statements “Fred and John are happy” and “Neither Fred nor John is happy?”

Q#04 Find the output of each of these combinatorial circuits.

- (a)



(b)



Q#05 Let $Q(x, y)$ be the statement “ x has sent an e-mail message to y ,” where the domain for both x and y consists of all students in your class. Express each of these quantifications in English.

a) $\exists x \exists y Q(x, y)$

b) $\exists x \forall y Q(x, y)$

c) $\forall x \exists y Q(x, y)$

d) $\exists y \forall x Q(x, y)$

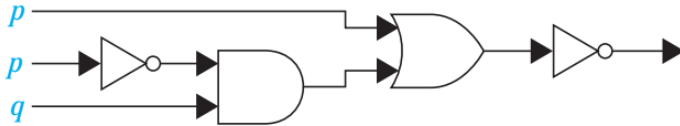
e) $\forall y \exists x Q(x, y)$

f) $\forall x \forall y Q(x, y)$

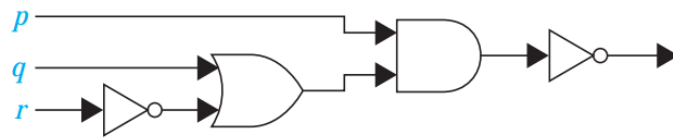
Group D

Q#01 Find the output of each of these combinatorial circuits.

(a)



(b)



Q#02 Use De Morgan's laws to find the negation of each of the following statements.

- a) Jan is rich and happy.
- b) Carlos will bicycle or run tomorrow.
- c) Mei walks or takes the bus to class.
- d) Ibrahim is smart and hard working.
- e) Kwame will take a job in industry or go to graduate school.
- f) Yoshiko knows Java and calculus.
- g) James is young and strong.
- h) Rita will move to Oregon or Washington.

Q#03 The truth value of the disjunction of two propositions in fuzzy logic is the maximum of the truth values of the two propositions. What are the truth values of the statements "Fred is happy, or John is happy" and "Fred is not happy, or John is not happy?"

Q#04 Logic Puzzles (Solve and understand the following puzzles from the book)

- (i) Example 7, page # 19, 7th edition
- (ii) Example 8, page # 20, 7th edition

Q#05 Show that the following are logically equivalent to tautology by developing a series of logical equivalences.

a) $\neg p \rightarrow (p \rightarrow q)$

b) $(p \wedge q) \rightarrow (p \rightarrow q)$

c) $\neg (p \rightarrow q) \rightarrow p$

d) $\neg (p \rightarrow q) \rightarrow \neg q$

e) $[\neg p \wedge (p \vee q)] \rightarrow q$

f) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

g) $[p \wedge (p \rightarrow q)] \rightarrow q$

h) $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

Group E

Q#01 Let p and q be the propositions

p : It is below freezing.

q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

- a) It is below freezing and snowing.
- b) It is below freezing but not snowing.
- c) It is not below freezing and it is not snowing.
- d) It is either snowing or below freezing (or both).
- e) If it is below freezing, it is also snowing.
- f) Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.
- g) That it is below freezing is necessary and sufficient for it to be snowing.

Q#02 Write each of these statements in the form “if p , then q ” in English.

- a) It is necessary to wash the boss's car to get promoted.
- b) Winds from the south imply a spring thaw.
- c) A sufficient condition for the warranty to be good is that you bought the computer less than a year ago.
- d) Willy gets caught whenever he cheats.
- e) You can access the website only if you pay a subscription fee.
- f) Getting elected follows from knowing the right people.
- g) Carol gets seasick whenever she is on a boat.
- h) It snows whenever the wind blows from the northeast.
- i) The apple trees will bloom if it stays warm for a week.
- j) That the Pistons win the championship implies that they beat the Lakers.
- k) It is necessary to walk 8 miles to get to the top of Long's Peak.
- l) To get tenure as a professor, it is sufficient to be world-famous.
- m) If you drive more than 400 miles, you will need to buy gasoline.
- n) Your guarantee is good only if you bought your CD player less than 90 days ago.

o) Jan will go swimming unless the water is too cold.

Q#03 A says “We are both knaves” and B says nothing. Exercises (i)–(viii) relate to inhabitants of an island on which there are three kinds of people: knights who always tell the truth, knaves who always lie, and spies (or called normals) who can either lie or tell the truth. You encounter three people, A, B, and C. You know one of these people is a knight, one is a knave, and one is a spy. Each of the three people knows the type of person each of other two is. For each of these situations, if possible, determine whether there is a unique solution and determine who the knave, knight, and spy are. When there is no unique solution, list all possible solutions or state that there are no solutions.

- (i) A says “C is the knave,” B says, “A is the knight,” and C says “I am the spy.”
- (ii) A says “I am the knight,” B says “I am the knave,” and C says “B is the knight.”
- (iii) A says “I am the knave,” B says “I am the knave,” and C says “I am the knave.”
- (iv) A says “I am the knight,” B says “A is telling the truth,” and C says “I am the spy.”
- (v) A says “I am the knight,” B says, “A is not the knave,” and C says “B is not the knave.”
- (vi) A says “I am the knight,” B says “I am the knight,” and C says “I am the knight.”
- (vii) A says “I am not the spy,” B says “I am not the spy,” and C says “A is the spy.”
- (viii) A says “I am not the spy,” B says “I am not the spy,” and C says “I am not the spy.”

Group F

Q#01 Let p, q, and r be the propositions

p : Grizzly bears have been seen in the area.

q : Hiking is safe on the trail.

r : Berries are ripe along the trail.

Write these propositions using p, q, and r and logical connectives (including negations).

- a) Berries are ripe along the trail, but grizzly bears have not been seen in the area.
- b) Grizzly bears have not been seen in the area and hiking on the trail is safe, but berries are ripe along the trail.
- c) If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.
- d) It is not safe to hike on the trail, but grizzly bears have not been seen in the area and the berries along the trail are ripe.
- e) For hiking on the trail to be safe, it is necessary but not sufficient that berries not be ripe along the trail and for grizzly bears not to have been seen in the area.
- f) Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.

Q#02 State the converse, contrapositive, and inverse of each of these conditional statements.

- a) If it snows today, I will ski tomorrow.
- b) I come to class whenever there is going to be a quiz.
- c) A positive integer is a prime only if it has no divisors other than 1 and itself.
- d) If it snows tonight, then I will stay at home.
- e) I go to the beach whenever it is a sunny summer day.
- f) When I stay up late, it is necessary that I sleep until noon.

Q#03

(i) Determine the truth value of each of these statements if the domain for all variables consists of all integers.

- a) $\forall n (n^2 \geq 0)$
- b) $\exists n (n^2 = 2)$
- c) $\forall n (n^2 \geq n)$

d) $\exists n (n^2 < 0)$

(ii) Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.

a) $\exists x (x^2 = 2)$

b) $\exists x (x^2 = -1)$

c) $\forall x (x^2 + 2 \geq 1)$

d) $\forall x (x^2 \neq x)$

(iii) Suppose that the domain of the propositional function $P(x)$ consists of the integers 0, 1, 2, 3, and 4. Write out each of these propositions using disjunctions, conjunctions, and negations.

a) $\exists x P(x)$

b) $\forall x P(x)$

c) $\exists x \neg P(x)$

d) $\forall x \neg P(x)$

e) $\neg \exists x P(x)$

f) $\neg \forall x P(x)$

Q#04 Explain Example#28 from book, page # 51, 7th edition.

Group G

Q#01 Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives.

- a) No one is perfect.
- b) Not everyone is perfect.
- c) All your friends are perfect.
- d) At least one of your friends is perfect.
- e) Everyone is your friend and is perfect.
- f) Not everybody is your friend or someone is not perfect.

Q#02

- (i) Explain Nested Quantifiers.
- (ii) Why the topic nested quantifiers in discrete structures is necessary for AI students to understand?
- (iii) What are the applications of nested quantifiers in AI?
- (iv) Give 3 examples of nested quantifiers related to AI.

Q#03 Translate these statements into English, where the domain for each variable consists of all real numbers.

- a) $\exists x \forall y (xy = y)$
- b) $\forall x \forall y (((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0))$
- c) $\forall x \forall y \exists z (x = y + z)$

Q#04 Let $T(x, y)$ mean that student x likes cuisine y , where the domain for x consists of all students at your school and the domain for y consists of all cuisines. Express each of these statements by a simple English sentence.

- a) $\neg T(\text{Abdallah Hussein}, \text{Japanese})$
- b) $\exists x T(x, \text{Korean}) \wedge \forall x T(x, \text{Mexican})$
- c) $\exists y (T(\text{Monique Arsenault}, y) \vee T(\text{Jay Johnson}, y))$
- d) $\forall x \forall z \exists y ((x \neq z) \rightarrow \neg(T(x, y) \wedge T(z, y)))$
- e) $\exists x \exists z \forall y (T(x, y) \leftrightarrow T(z, y))$
- f) $\forall x \forall z \exists y (T(x, y) \leftrightarrow T(z, y))$

Q#05 Let $I(x)$ be the statement “ x has an Internet connection” and $C(x, y)$ be the statement “ x and y have chatted over the Internet,” where the domain for the variables x and y consists of all students in your class. Use quantifiers to express each of these statements.

- a) Jerry does not have an Internet connection.
- b) Rachel has not chatted over the Internet with Chelsea.
- c) Jan and Sharon have never chatted over the Internet.
- d) No one in the class has chatted with Bob.
- e) Sanjay has chatted with everyone except Joseph.
- f) Someone in your class does not have an Internet connection.
- g) Not everyone in your class has an Internet connection.
- h) Exactly one student in your class has an Internet connection.
- i) Everyone except one student in your class has an Internet connection.
- j) Everyone in your class with an Internet connection has chatted over the Internet with at least one other student in your class.
- k) Someone in your class has an Internet connection but has not chatted with anyone else in your class.
- l) There are two students in your class who have not chatted with each other over the Internet.
- m) There is a student in your class who has chatted with everyone in your class over the Internet.
- n) There are at least two students in your class who have not chatted with the same person in your class.
- o) There are two students in the class who between them have chatted with everyone else in the class.